

UCS645 – Parallel & Distributed Computing

Assignment 4 – Introduction to MPI

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1. Introduction

Message Passing Interface (MPI) is a standardized library used for parallel programming in distributed memory systems. This assignment implements and analyzes four MPI programs and evaluates their performance using execution time, speedup, and efficiency metrics.

2. Performance Metrics

Execution Time (T_p): Time taken with p processes. Speedup (S_p): $S_p = T_1 / T_p$. Efficiency (E_p): $E_p = S_p / p$. Communication Time: Time spent in MPI functions. Computation Time: Time spent in arithmetic operations. Scalability: How performance changes with increasing processes.

Exercise 1 – Ring Communication (Execution Time)

Processes	T_p (sec)
1	0.000000
2	0.000038
4	0.000063
8	0.009732

Exercise 2 – Parallel Array Sum

Processes	T_p (sec)	Speedup	Efficiency
1	0.007877	1.00	1.00
2	0.008757	0.90	0.45
4	0.003519	2.24	0.56
8	0.003562	2.21	0.28

Exercise 3 – Global Max & Min

Processes	Tp (sec)	Speedup	Efficiency
1	0.020003	1.00	1.00
2	0.008955	2.23	1.11
4	0.006200	3.23	0.81
8	0.005782	3.46	0.43

Exercise 4 – Parallel Dot Product

Processes	Tp (sec)	Speedup	Efficiency
1	0.182622	1.00	1.00
2	0.156431	1.17	0.59
4	0.110868	1.65	0.41
8	0.258610	0.71	0.09

Final Conclusion

The performance of MPI programs depends strongly on the ratio of computation to communication. Communication-dominant programs show poor scalability, while computation-heavy programs achieve better speedup. Oversubscription and communication overhead limit efficiency at higher process counts.